Program structures

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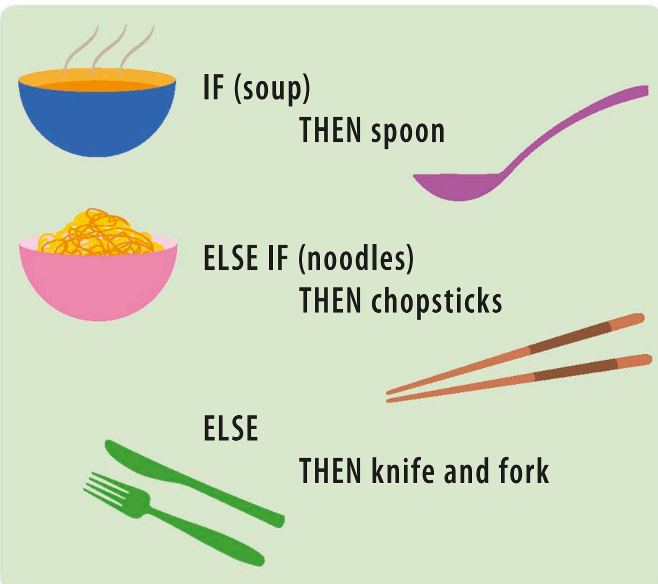
A program can be described as a collection of possible sequences of instructions. To determine which lines of code to execute, programs use control structures such as branches and loops.

Branching

A boat sailing towards a branch in the river can either go left, or right. It can't sail down both streams at the same time. Similarly, the IF-THEN-ELSE control structure sends the program down a single branch of possibilities and ignores code in other branches. The choice of path typically depends on the piece of data stored in a variable.



△ **Branching structure**
A spoon, chopsticks, and knife and fork aren't all required for a single meal. Using a branching structure in a program forces the user to pick the best option, depending on the circumstances.

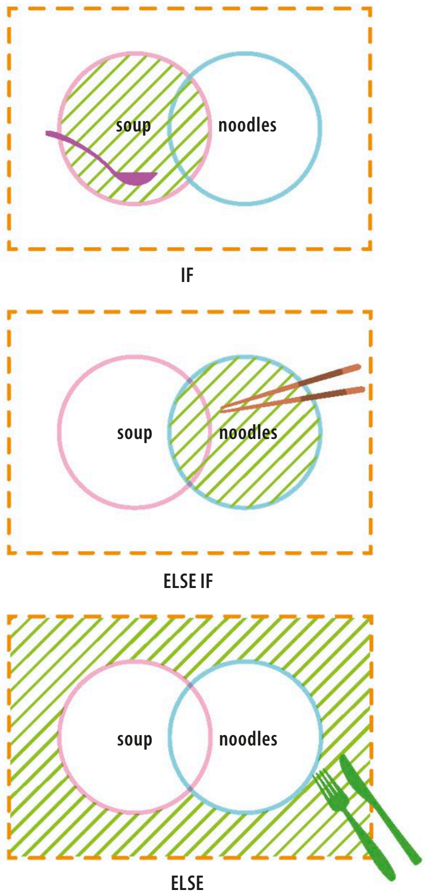
IN DEPTH

IF-THEN-ELSE

"If", "else if" (elif), and "else" are common programming keywords used to make decisions. "If" checks an initial condition, such as asking, "Is the sky blue?" If that condition is false, "else if" checks an alternative, such as, "Is the sky purple?" The program defaults to "else" if none of those conditions can be met.

Grouping data

The IF-THEN-ELSE control structure is used as a way to sort data into groups. Programmers can also write customized code to manipulate each group differently. The ELSE statement is a "catch-all" group for data that doesn't fit into the previous categories.



△ **Boolean algebra**
Venn diagrams can be used to represent how data is separated into groups. Boolean algebra can also be used to create more complex groups, such as soup AND noodles.